

Tune plan — same 13 steps, more H/s

This Mac, measured 11 Sep 2026. Do not rewrite RandomX 4.6.2. A changed step yields a different R; the pool rejects it. Better miner = the same recipe, more times per second.

0. Rule

XMRRig already JIT-compiles the 13-step loop to ARM and uses hardware AES. The improvement surface is four things: thread count, keep the 2 GiB dataset uncompressed, run one miner, keep the Mac awake. Linux 1 GB hugepages are **not installable on macOS** and are not in this plan.

1. This machine (inspected)

Fact	Value
Chip	Apple M4, arm64
Cores	10 logical = 4P + 6E
RAM	16 GiB
At inspect	free ~2.5 GiB · compressor ~3.0 GiB · wired ~2.4 GiB · swap has been used historically
XMRRig process	none
minerv3 install	not present (~/.Library/Application Support/minerv3 missing)
Binary used for bench	Desktop/XMR-Miner-Portable/bin/xmrig (6.26.0, signed ad-hoc)
Hugepages	macOS has none. API hugepages: [0, ...] is normal.

2. Thread sweep (offline --bench, no pool)

Each RandomX thread holds a 2 MiB scratchpad. Throughput is how many of those stay hot, not “use every core.” Sweep: fast mode, warmup 70 s, max of 6 API samples. Thread list 4–10 (this chip has 10 logical cores; the script default 3–8 would have missed the top).

Threads	Hashrate	Note
4	2,744.8 H/s	P-cluster only
5	2,828.9 H/s	
6	3,465.6 H/s	
7	3,738.8 H/s	
8	3,807.0 H/s	M2 peak was 3,212 at 8
9	4,092.5 H/s	
10	4,204.4 H/s	best on this M4

Decision: 10 threads. Climb is monotonic. Unlike the source M2 (peak at all 8 cores, then 9–10 lost), this M4 still gains through 10. After you run `install.sh`, it already sets `--threads` and `--randomx-init` to logical cores = 10. Leave them. Do not copy the M2 “8 is best” number onto this machine.

3. Hugepages — cannot install (macOS)

Linux --randomx-1gb-pages maps the dataset on 1 GiB pages so TLB misses stay low. macOS has no huge-page allocator. There is nothing to install, no sysctl, no brew package. XMRig will print hugepages [0, ...]. That is expected, not a defect.

The macOS substitute is the next section: keep the 2,080 MiB dataset **resident and uncompressed**. If the compressor eats it, step 7 (DRAM XOR) becomes a decompress. That is the silent H/s cliff, the same class of problem hugepages prevent on Linux.

4. Protect the 2 GiB book

Fast mode needs ~2,080 MiB dataset + 2 MiB × threads. At 10 threads that is ~2,100 MiB. 16 GiB is enough **if** the rest of the system leaves that slab alone.

Do	Why
Run only one xmrigh	A second process allocates a second 2 GiB. On the M2 that took 3,100 H/s to 240 H/s.
Keep ~3+ GiB free while mining	At inspect, free was 2.5 GiB and compressor already held 3.0 GiB. Start mining with browsers/IDEs light, then watch.
If H/s drops suddenly, check compressor	memory_pressure and vm_stat “Pages used by compressor”. A jump there with flat CPU is the dataset being packed.
Do not put the binary in Desktop/Downloads for launchd	TCC: launchd cannot prompt. The job hangs in dyld at 0% CPU. install.sh puts it in Application Support for this reason.
Leave --randomx-mode=fast	Light mode is the same R, much slower (recomputes items). You have 16 GiB; use fast.

```
# while mining:
memory_pressure
vm_stat | grep compressor
# hashrate (after install): curl -s http://127.0.0.1:18088/2/summary
```

5. One miner

install.sh and minerctl.sh start already refuse if xmrigh is running. Keep it that way. Do not launch the Desktop app and a Terminal copy. Do not run the HTML animation’s “miner” — there isn’t one; the desk is a picture.

6. Keep the Mac awake

After install, the launchd job wraps xmrigh in caffeinate -i and ProcessType=Standard (not Background — Background parks work on E-cores and guts RandomX). Click-to-start only: the plist is not in LaunchAgents, so a reboot does not mine until you click the app. That is intentional. Lid-close: caffeinate -i prevents idle sleep while the job is loaded; clamshell on power still works. Unplug + lid close can still sleep on some Macs — leave it open or use a dummy HDMI if you need it closed.

7. What install.sh will do on this Mac (not run yet)

minerv3 is not installed. When you run Desktop/XMR-Miner-Portable/install.sh it will detect 10 cores / 16 GB and write --threads=10 --randomx-init=10. That matches the sweep. Confirm the prompt shows Threads: 10 before typing y.

```
# after install, threads and randomx-init must both be 10
# file: ~/Library/Application Support/minerv3/com.minerv3.xmrigh.plist
```

Do not start mining from this document. Start is: double-click XMR Miner.app after install, or minerctl.sh start.

8. Expected speed

Machine	Threads	H/s (offline bench)
Source M2 (pack README)	8	~3,212
This M4 (this sweep)	10	4,204

Live pool H/s will sit a little under the bench (job changes, network). If live is far below ~4,000 on this M4 at 10 threads, look at compressor and a second xmrig before touching the 13 steps.

9. Checklist (after you choose to install)

#	Check
1	One xmrig only (pgrep -x xmrig).
2	plist --threads=10 and --randomx-init=10.
3	--randomx-mode=fast. Algorithm rx/0.
4	Job is caffeinate -i + ProcessType Standard.
5	Hugepages [0] — ignore.
6	API H/s near 4.2k after ~1 min warmup. Sudden drop → compressor / RAM, not a new hash.
7	Shares accepted. Rejects mean wrong R or stale job, not thread count.
8	Do not rewrite 4.6.2.

Sweep log: /tmp/xmr_bench_sweep_m4.log. Binary: Desktop/XMR-Miner-Portable/bin/xmrig. Companion: Desktop/RandomX-mining-audit.pdf (the 13 steps this plan refuses to change).